

Raw Photos

Agajo, Aleck Danielle V.

BSIT 2-1N

Object Oriented Programming

> Part A

VARIABLE NAME	DATA TYPE	VARIABLE TYPE	STORED IN
name	String	Instance Variable (field)	Heap (made Student object)
grade	int	Instance Variable (field)	Heap (inside Student object)
s1	Student	Reference Variable (local)	Stack
s2	Student	Reference Variable (local)	Stack
s3	Student	Reference Variable (local)	Stack
x1	String	Parameter (local variable)	Stack
g	int	Parameter (local variable)	Stack
args	String[]	Parameter (main method)	Stack

> Part 4

STATEMENT	WHAT HAPPENS
1) Student s1 = new Student ("John", 85);	a new Student object is created // s1 stores the reference (address) to new object
2) Student s2 = new Student ("Maria", 90);	another new Student object is created // s2 stores reference to this new object
3) Student s3 = s2;	no new object is created // s3 copies the reference found in s2 // points to the same object (object 5)
4) s1.grade = 85;	updates grade of Object A
5) s3.name = "Ana";	s3 points to Object B // name of Object B changes from "Maria" to "Ana"

MEMORY IMPACT

- 1) STACK: s1's reference to Object A // HEAP: Object A is { name = "John", grade = 85 }
- 2) STACK: s1's Object A - s2's Object B // HEAP: Object A is { John, 85 } - Object B is { Maria, 90 }
- 3) STACK: s1's Object A - s2's Object B - s3's Object B // HEAP: unchanged/unaffected
- 4) STACK: unchanged/unaffected // Object A is { John, 85 } - Object B is { Maria, 90 }
- 5) STACK: unchanged/unaffected // Object A is { John, 85 } - Object B is { Ana, 90 }

> Part C

John 9:5 (light + dark + blue) 9:5

Ann 9:10 (light + dark + blue) 9:10

Ann 9:10

Agarwal, Meekha Danielle V.

BSIT 2-14

Object Oriented Programming

> Part A

Variable Name	Data Type	Variable Type	Stored In
name	String	Instance Variable (field)	Heap (inside Student object)
grade	int	Instance Variable (field)	Heap (inside Student object)
s1	Student	Reference Variable (local)	Stack
s2	Student	Reference Variable (local)	Stack
s3	Student	Reference Variable (local)	Stack
x1	String	Parameter (local variable)	Stack
g	int	Parameter (local variable)	Stack
args	String[]	Parameter (main method)	Stack

> Part B

Statement	What Happens
1) Student s1 = new Student ("John", 85);	1) new Student object is created // s1 stores the reference (address) to this new object
2) Student s2 = new Student ("Maria", 90);	2) another new Student object is created // s2 stores reference to this new object
3) Student s3 = s2;	3) no new object is created // s3 copies the reference stored in s2 // points to the same object (object 2)
4) s1.grade = 85;	4) updates grade of Object A
5) s3.name = "Ana";	5) s3 points to Object B // name of Object B changes from "Maria" to "Ana"

Memory Impact

1) STACK: s1's reference to Object A // HEAP: Object A's { name = "John", grade = 85 }	
2) STACK: s1's Object A - s2's Object B // HEAP: Object A's { John, 85 } - Object B's { Maria, 90 }	
3) STACK: s1's Object A - s2's Object B - s3's Object B // HEAP: unchanged/unmodified	
4) STACK: unchanged/unmodified // Object A's { John, 85 } - Object B's { Maria, 90 }	
5) STACK: unchanged/unmodified // Object A's { John, 85 } - Object B's { Ana, 90 }	

> Part C

John 9:15 (lighter shade) 9:15

Ann 9:10 (lighter shade) 9:10

Ann 9:10